Department of Engineering Science
School of Science & Technology



CES 440 Data Communications
Ch.4 The MAC Sublayer: Part II

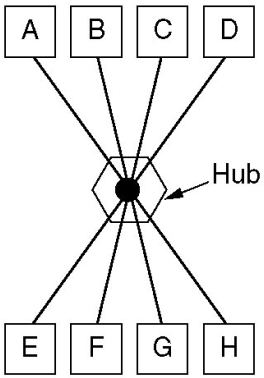
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OUTLINE

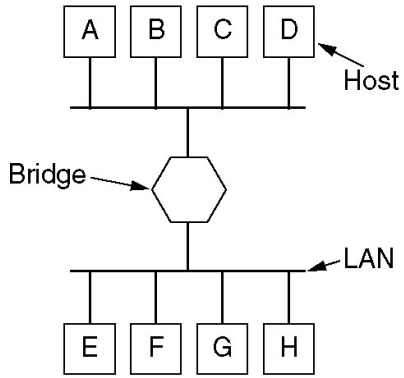
- Functions of MAC Sublayer
- MAC Protocols
- Ethernet (IEEE 802.3)
- **Data Link Layer Switching**
- Wireless LAN
- Broadband Wireless
- Bluetooth

DATA LINK LAYER SWITCHING: TERMINOLOGIES

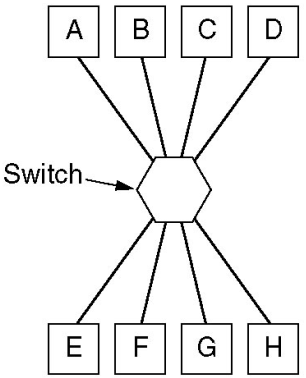
- **Terminologies: Repeater, Hub, Bridge, Switch, Router, Gateway**
  - Repeater: Analog device amplifying signal (Physical Layer)
    - Doesn't care about frame contents: only cares voltage
  - Hub: Connecting multiple hosts like a shared cable (Physical Layer)
    - Doesn't care about frame contents
  - Bridge: Connecting two or more LANs (Data Link Layer)
    - Examine the frame headers, route based on frame address (e.g. MAC address)
    - Different LANs can have different data link layer
  - Switch: Connecting two or more hosts (Data Link Layer)
    - Examine the frame headers, route based on frame address



(a)



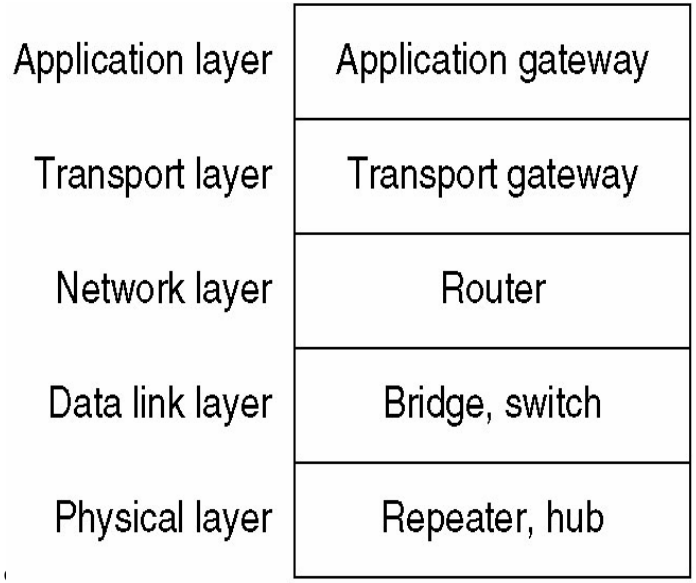
(b)



(c)

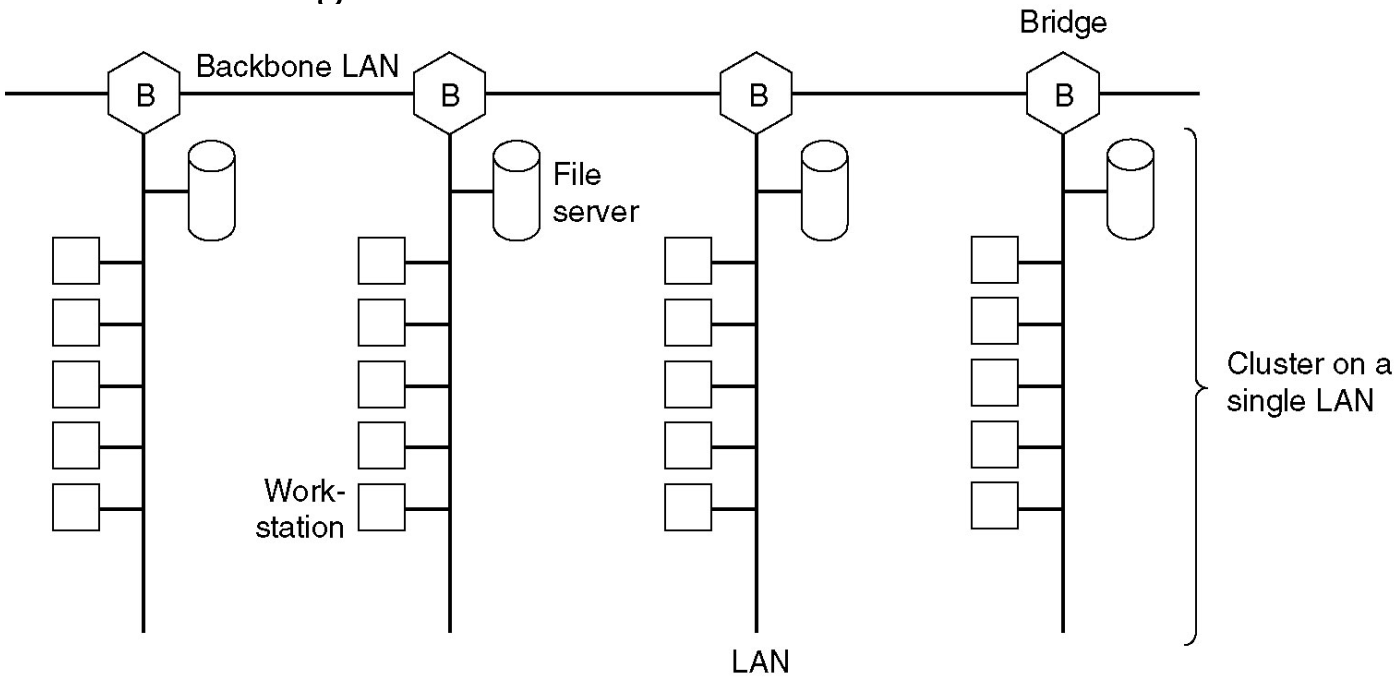
DATA LINK LAYER SWITCHING: TERMINOLOGIES

- **Terminologies (Cont'd)**
  - Router: Routing packets based on network layer header
    - E.g. routing based on IP address.
  - Transport Gateway: Connecting two hosts with different transport layer protocols
    - E.g. One host using TCP/IP protocol talk with another one using ATM transport protocol.
  - Application Gateway: Translating data from one format to another.
    - E.g. E-mail gateway: translate regular email message into SMS message for mobile phones.



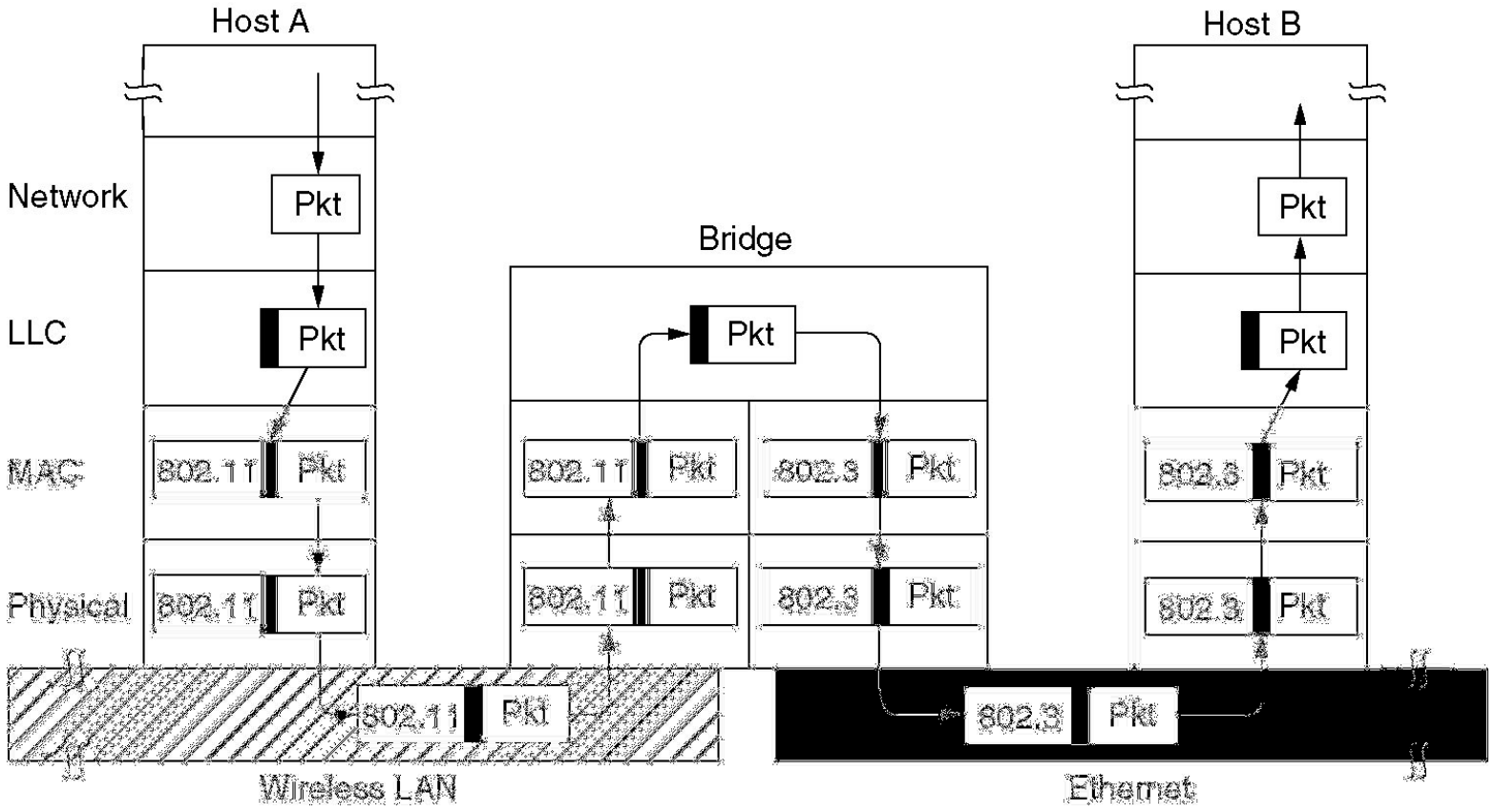
DATA LINK LAYER SWITCHING: BRIDGES

- **Bridges: Connecting two or more LANs to handle a total load higher than a single LAN**
  - Should be transparent to users: higher layers should not notice the existence of bridges



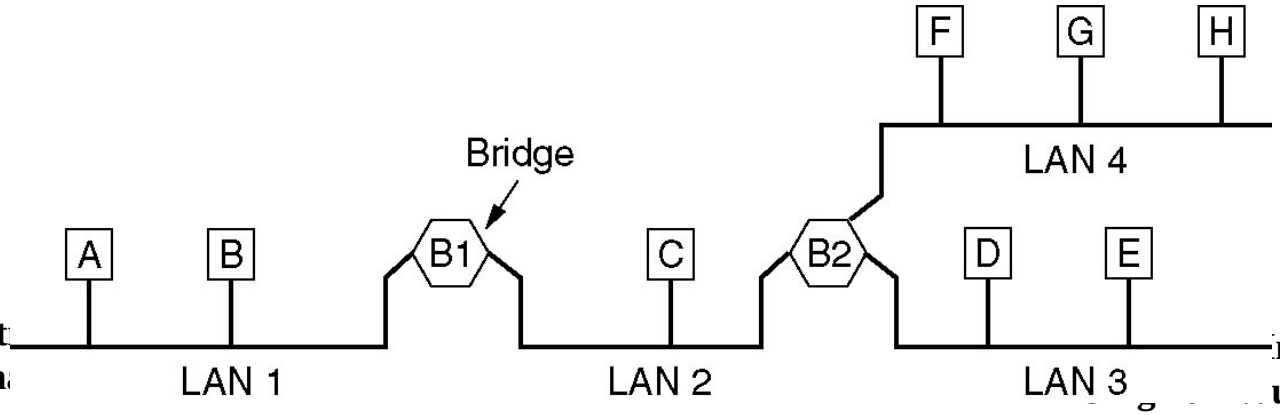
SWITCHING: BRIDGES FROM 802.X TO 802.Y

- **Example: 802.11 LAN connecting with 802.3 LAN**



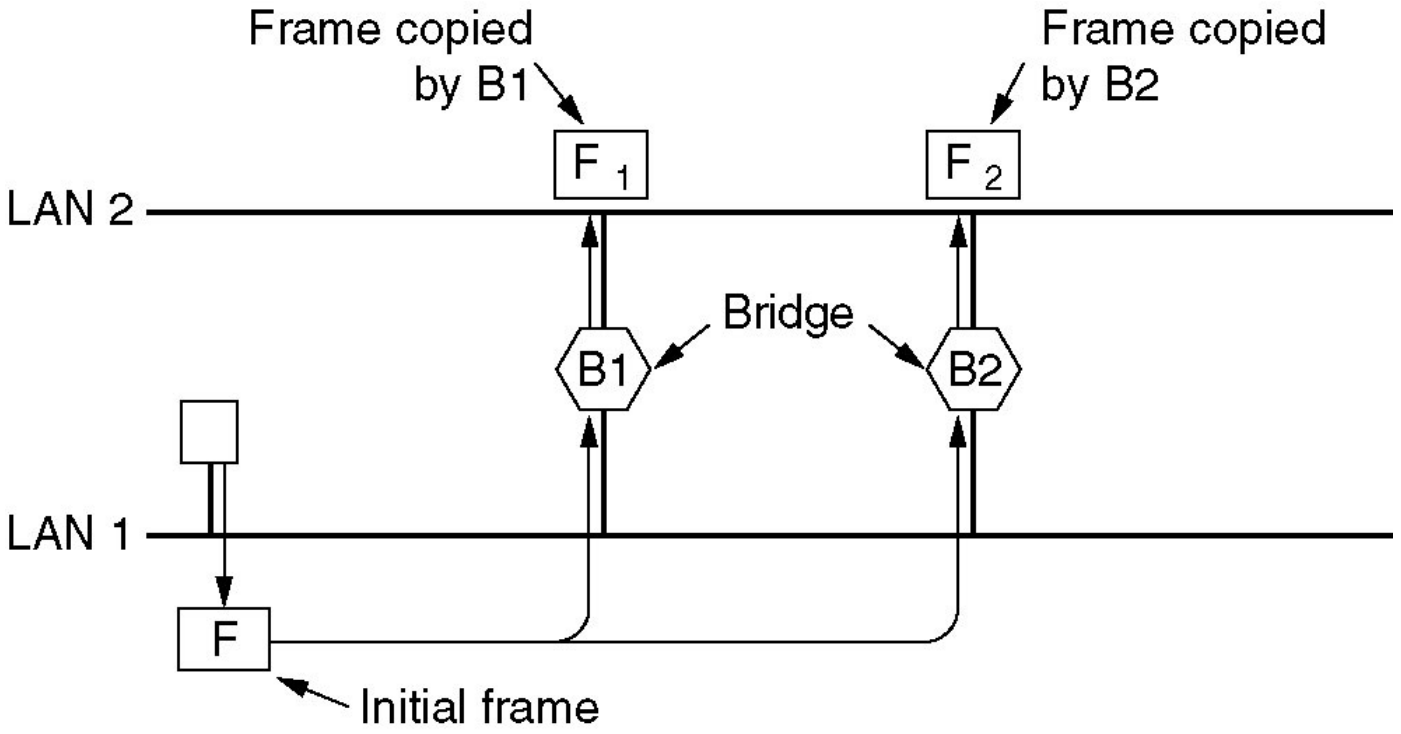
SWITCHING: LOCAL INTERNETWORKING

- **How bridge forward frame to its destination? ➔ Destination Table with Backward Learning**
  - Each bridge maintains a hash table with entry: (host, LAN#)
    - E.g. on B1: (A, LAN 1), (C, LAN 2), (F, LAN 2)
  - When bridges powered up, the table is empty, and it works at promiscuous mode: send frame to all the output lines except the incoming line ➔ flooding.
  - It will fill in the table entries by examining the source address of incoming frames ➔ backward learning
    - E.g. receive a frame from F on LAN 2, new entry: (F, LAN 2)
  - Bridges will purge entries more than a few minutes old ➔ computers might be moved to other locations or removed.
  - For frame with unknown destination, use flooding.



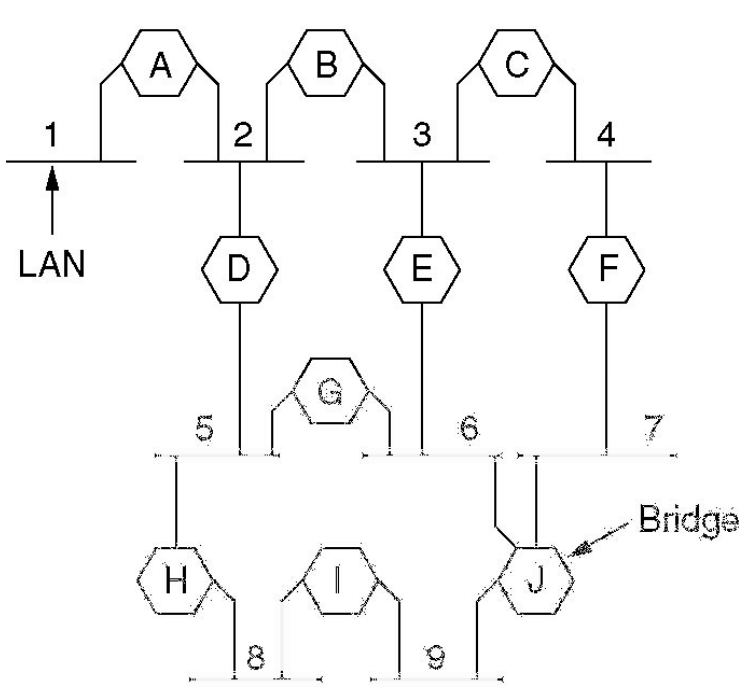
SWITCHING: SPANNING TREE BRIDGES

- **What is spanning tree protocol?**
  - An algorithm used to avoid the loops in LAN connections.

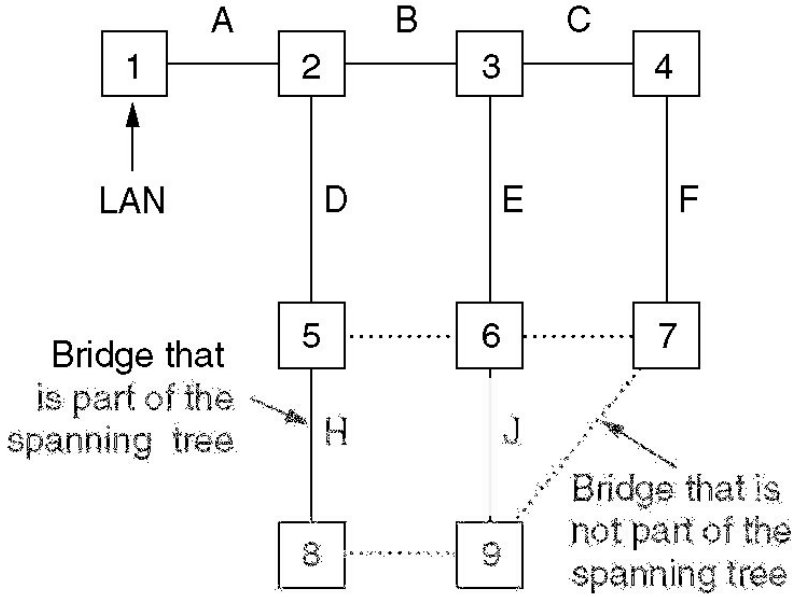


SWITCHING: SPANNING TREE BRIDGES

- **How does spanning tree work?**
  - Bridges negotiate with each other to build a spanning tree topology (no loop) by omitting some of the redundant bridges.



(a)



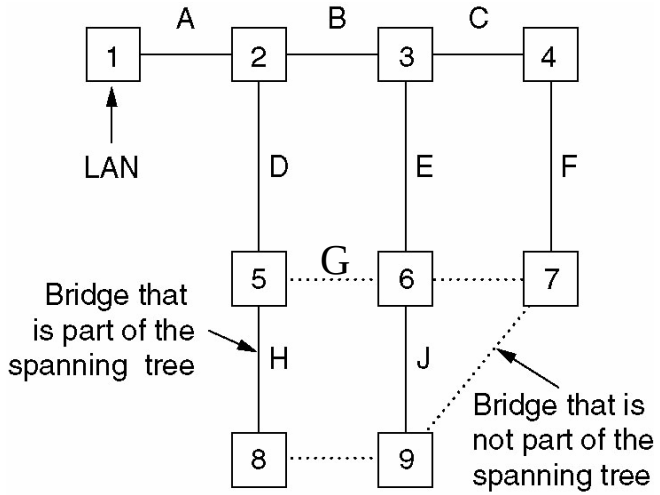
(b)

SWITCHING: SPANNING TREE BRIDGES

- **How does spanning tree work (cont'd)?**
  - The algorithm has been standardized in IEEE 802.1D
  - Initially, bridges need to choose one bridge as root
    - All bridges broadcast its serial# (unique). The one with the smallest is going to be the root.
  - Next, a tree of shortest paths from the root to every bridge and LAN is constructed.
    - When there are two parallel connections between to LANs, the one with the shorter path will survive.
    - E.g. 1 – 2 – 5 (survivor),

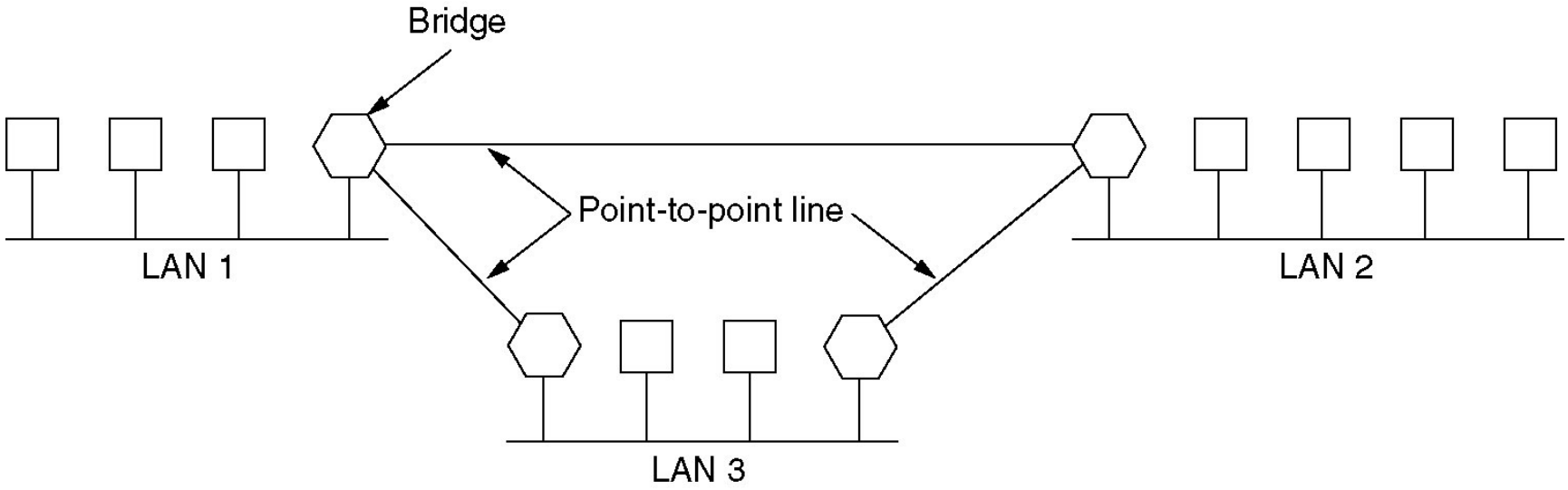
      1 – 2 – 3 – 6 – 5

      1 – 2 – 3 – 4 – 7 – 6 – 5



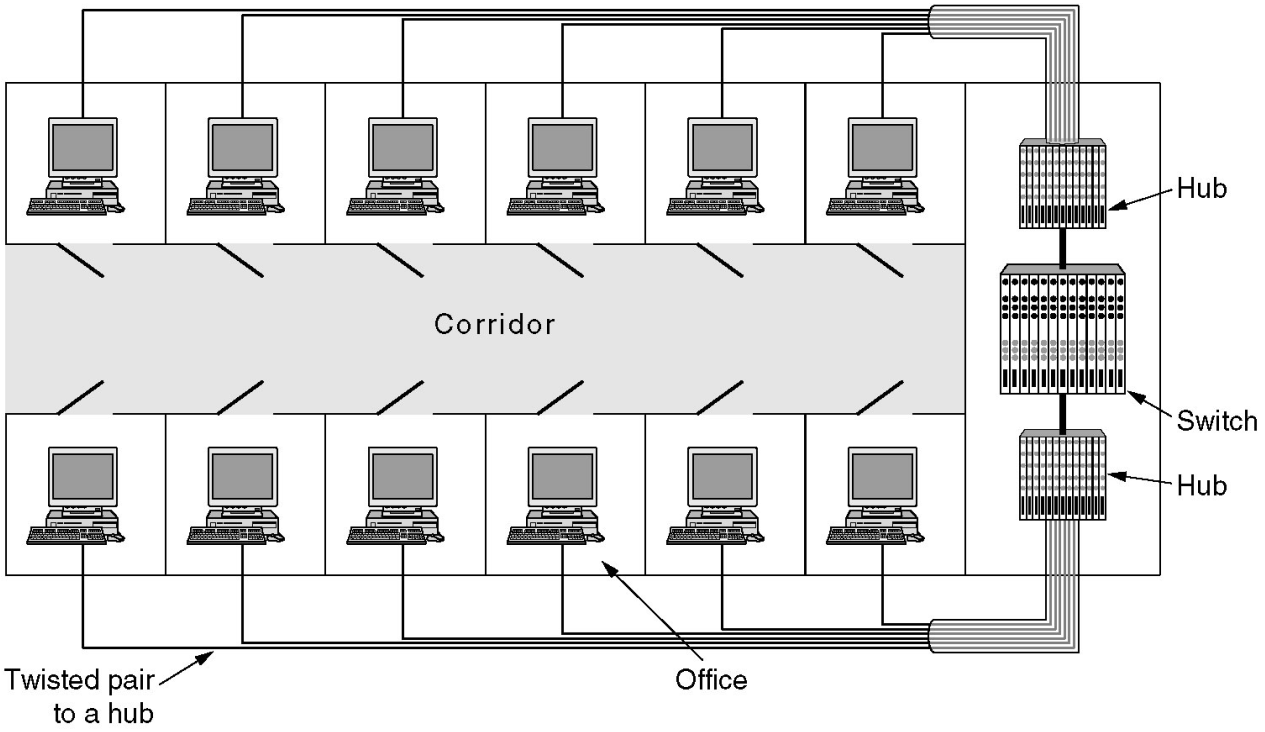
SWITCHING: REMOTE BRIDGES

- **Remote LANs can be interconnected by remote bridges and point-to-point lines.**
  - In the point-to-point line, two protocols can be used
    - 1. PPP, put the MAC frame in the payload field of PPP frame
    - 2. Strip off the MAC header, re-encapsulate the frame with the PPP protocol.



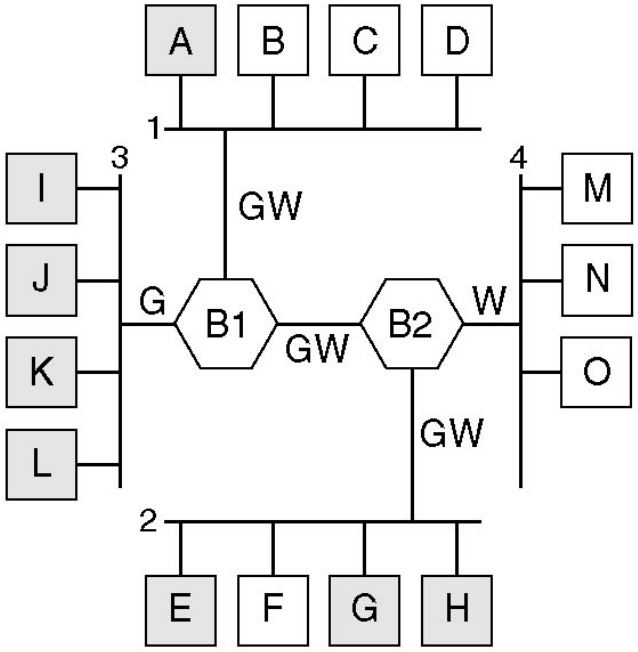
SWITCHING: VIRTUAL LAN (VLAN)

- **Why VLAN? ➔ Geographical Structure v.s. Logical Structure**
  - LANs are usually grouped based on geographical locations.
  - Users want LANs organized based on logical structures
    - E.g. research, patents, accounting might not want to share some information with each other, but they want the ability of broadcasting.

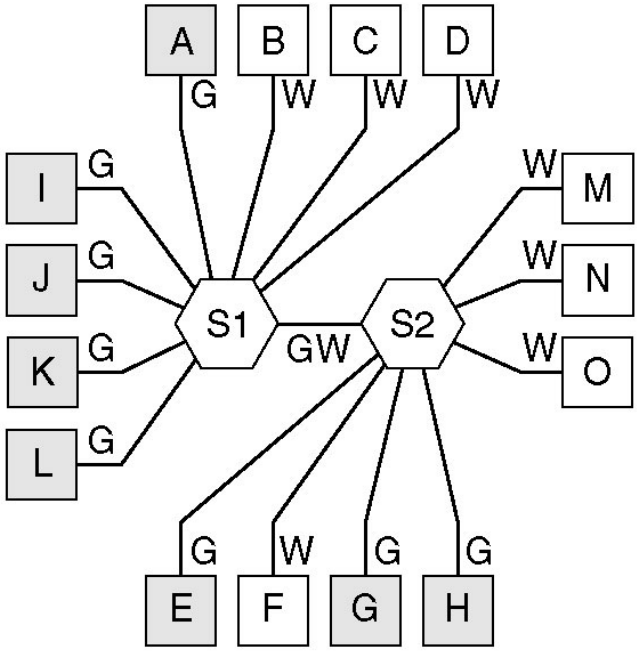


SWITCHING: VLAN

- **VLAN**
  - Use specially designed software to reorganize the structure of LANs.
  - The software is embedded in VLAN-aware switches.
  - Network administrator determines how many VLANs are needed and which machine is on which VLAN by configuring the switches/bridges.



(a)



(b)

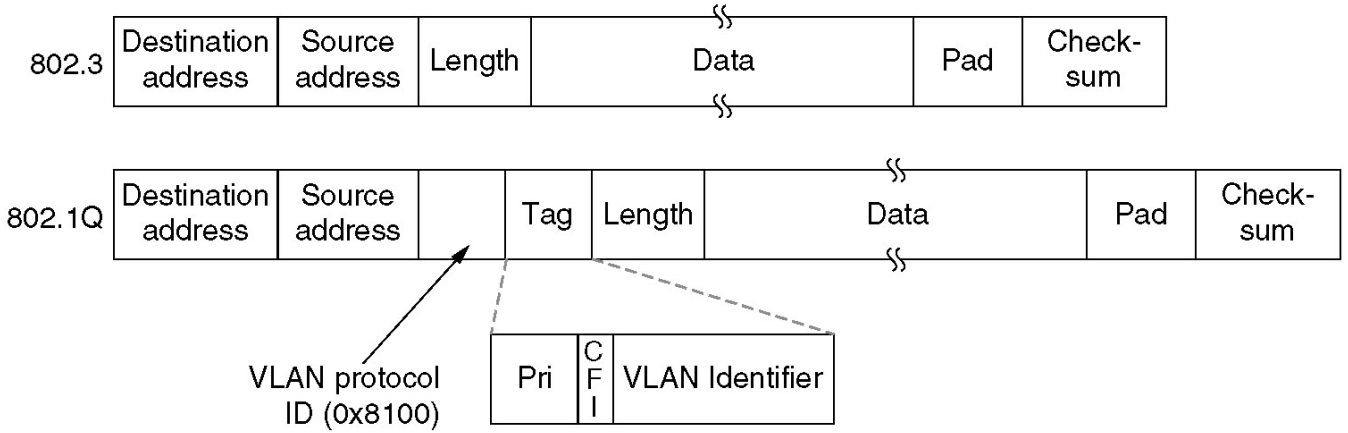
Two VLANs: Grey and White

SWITCHING: VLAN

- **Configuration table**
  - Every bridge/switch has a configuration table with entries (host, port #, VLAN id)
  - This table can be manually set up by network administrator based on logical structure.
- **How to find out a frame belonging to which VLAN?**
  - Use Port# of the switch
    - Configuration table entry (port#, VLAN id)
  - Use MAC address.
    - Configuration table entry (MAC address, VLAN id)
  - Use IP address or other network layer protocol address
    - Configuration table entry (IP address, VLAN id)
    - Violate the rule of layer independency.
    - Not widely used.

SWITCHING: VLAN

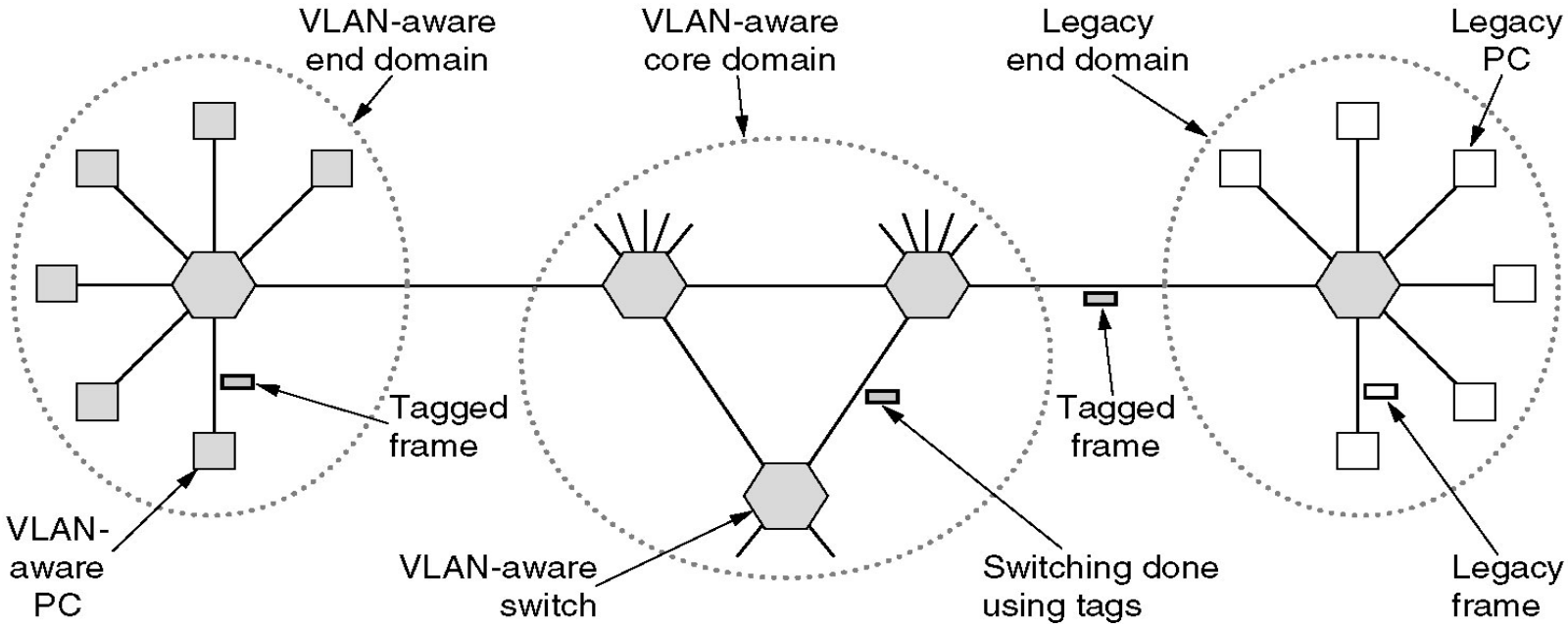
- **IEEE 802.1Q**
  - Published in 1998 to enable Ethernet with VLAN capability
  - Add a new VLAN field in Ethernet header!!!
    - Do we need to throw away the existing Ethernet card?
    - If not, who generate this new VLAN field?
  - Frame structure



Pri: priority, has nothing to do with VLAN
CFI: indicate 802.5 frame in payload, has nothing to do with VLAN

SWITCHING: VLAN

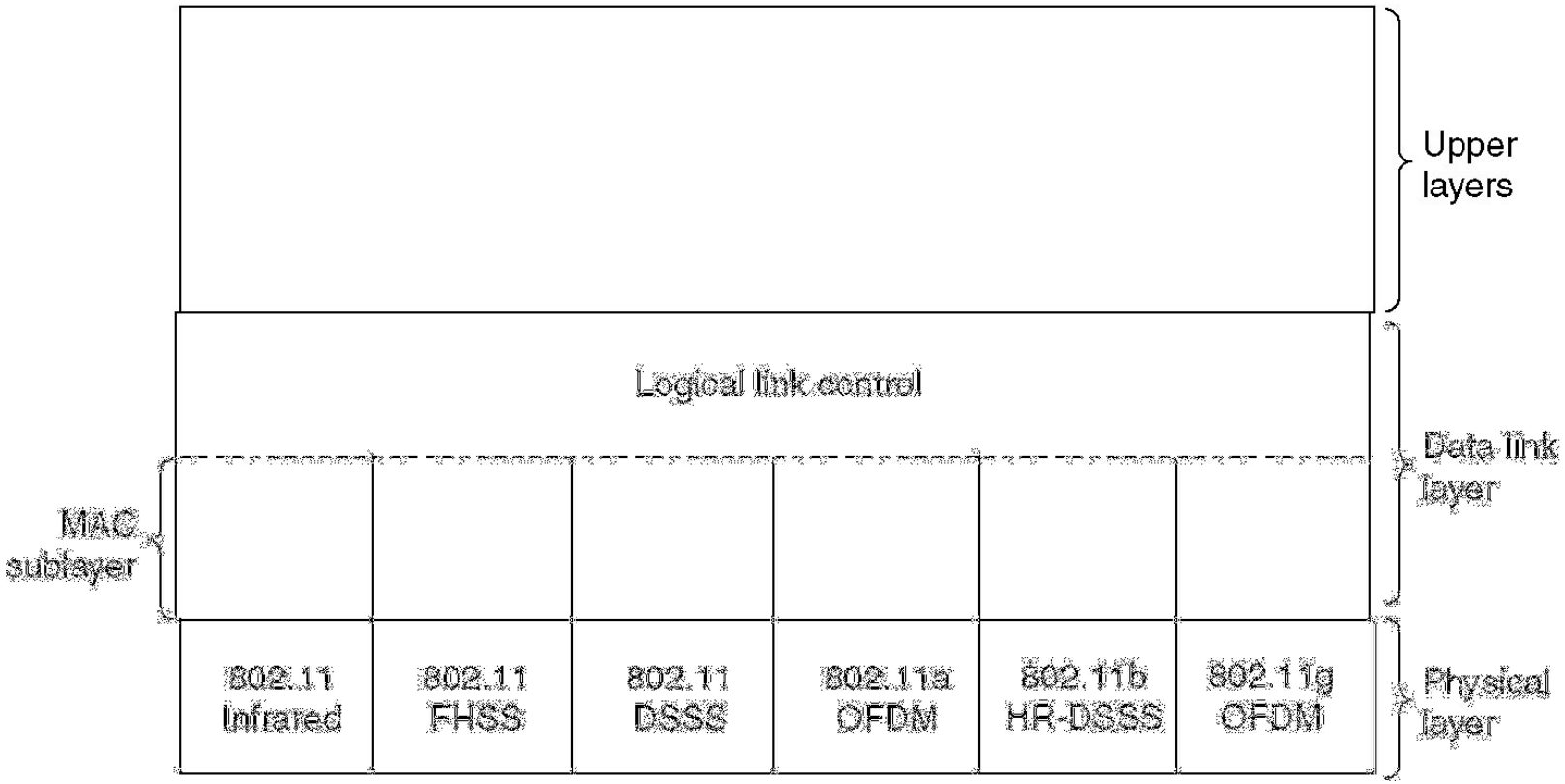
- **What if the host is not 802.1Q compatible?**
  - The first bridge adds the VLAN field in the incoming frame
  - The last bridge strips off the VLAN field in the outgoing frame.
  - 802.1Q hosts and non-802.1Q hosts can operate in the same network!



OUTLINE

- Functions of MAC Sublayer
- MAC Protocols
- Ethernet (IEEE 802.3)
- Data Link Layer Switching
- **Wireless LAN**
- Broadband Wireless
- Bluetooth

WIRELESS LAN (802.11): PROTOCOL STACK



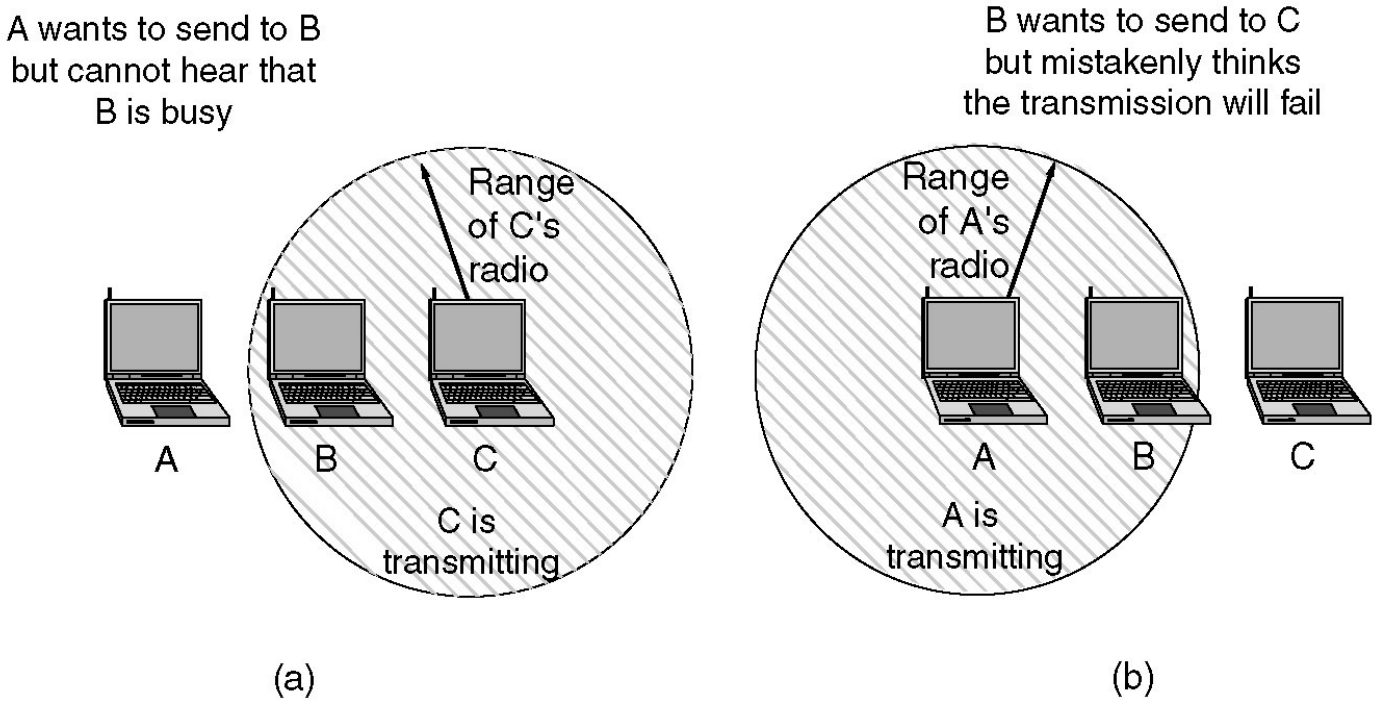
IEEE 802.11: PHYSICAL LAYER

- **Infrared**
  - 1Mbps, 2Mbps.
  - Not popular.
- **Frequency Hopping Spread Spectrum (FHSS)**
  - 1Mbps, 2Mbps
  - 79 channels (1MHz/channel), Operating at ISM 2.4GHz
  - During Tx, continuously hops from one channel to another
    - A pseudorandom number generator determines which channel to hop next ➔ good security! (widely used in military applications)
- **Direct Sequence Spread Spectrum (DSSS)**
  - Similar to CDMA.
  - 1Mbps, 2Mbps
  - 11 chips/bit.

IEEE 802.11: PHYSICAL LAYER

- **Orthogonal Frequency Division Multiplexing (OFDM)**
  - Similar to DMT used in ADSL:
    - spectrum divided into multiple smaller channels.
  - IEEE 802.11a:
    - Operating at 5GHz ISM band, up to 54Mbps.
  - IEEE 802.11g:
    - Operating at 2.4GHz ISM band, up to 54Mbps.
- **High Rate Direct Sequence Spread Spectrum (HR-DSSS)**
  - Similar to CDMA
  - IEEE 802.11b
  - Operating at 2.4GHz ISM band
  - Support 1Mbps, 2Mbps, 5.5Mbps, and 11Mbps

IEEE 802.11: MAC SUBLAYER



Hidden Station Problem

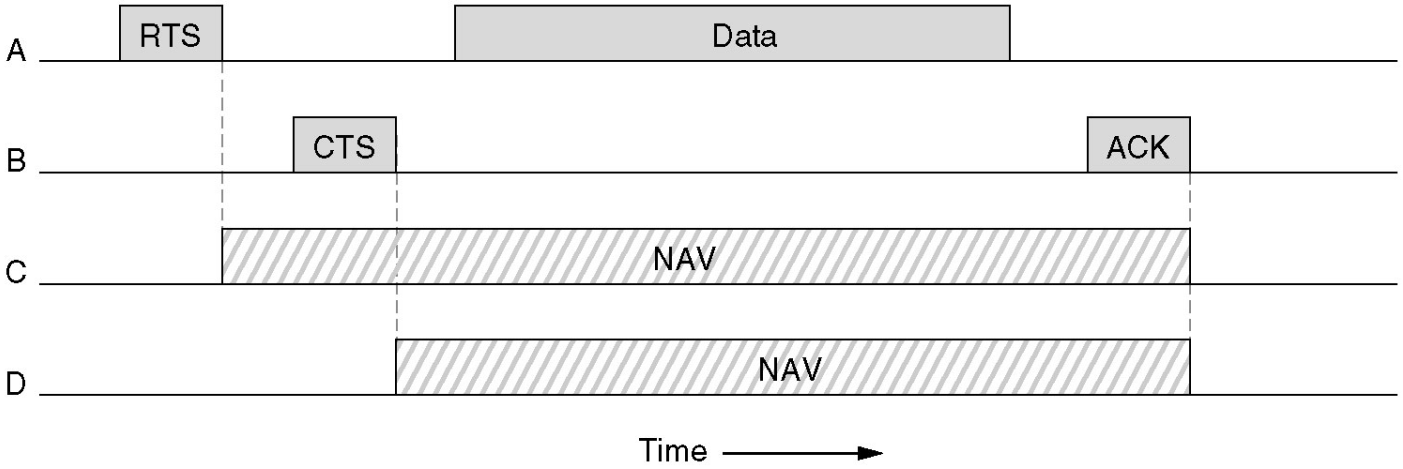
Exposed Station Problem

IEEE 802.11: MAC SUBLAYER

- **Two modes of operation**
  - DCF (Distributed Coordination Function)
    - No central control, all stations compete for channel ➔similar to Ethernet.
    - Need to address the hidden station and exposed station problem.
  - PCF (Point Coordination Function)
    - Base station controls all activity in the cell ➔ No collision.
    - Base station sends out beacon frame periodically
      - Consists system parameters
      - Invites new station to sign up. Once a new station signed up, it's guaranteed to have a portion of the channel.

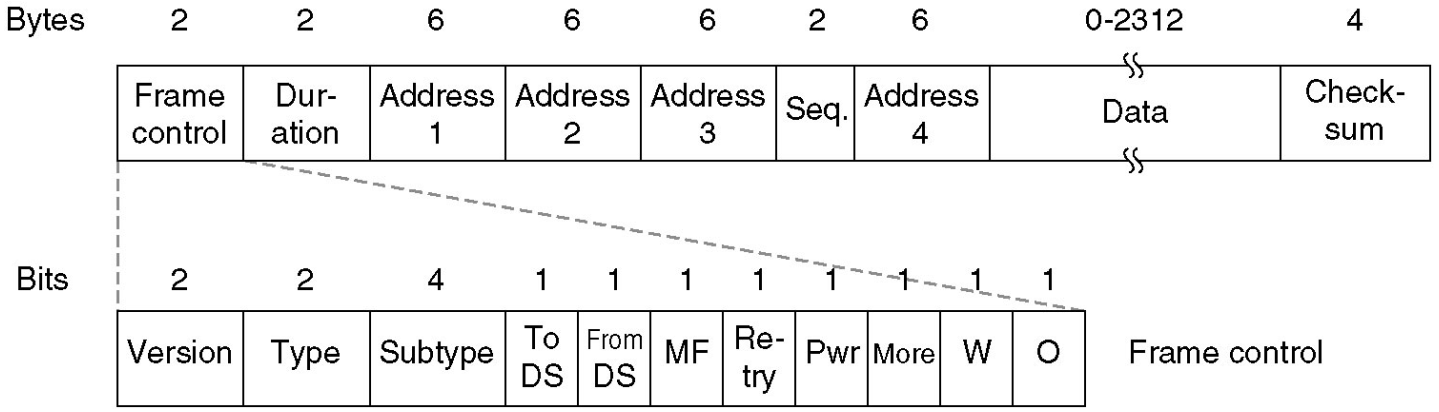
IEEE 802.11: MAC SUBLAYER

- **DCF**
  - Method 1:
    - Sensing the channel before Tx.
    - Doesn't sense the channel during Tx (unlike CD of Ethernet!)
    - If collision, binary exponential backoff
  - Method 2: CSMA/CA
    - RTS (Request To Send), CTS (Clear To Send)



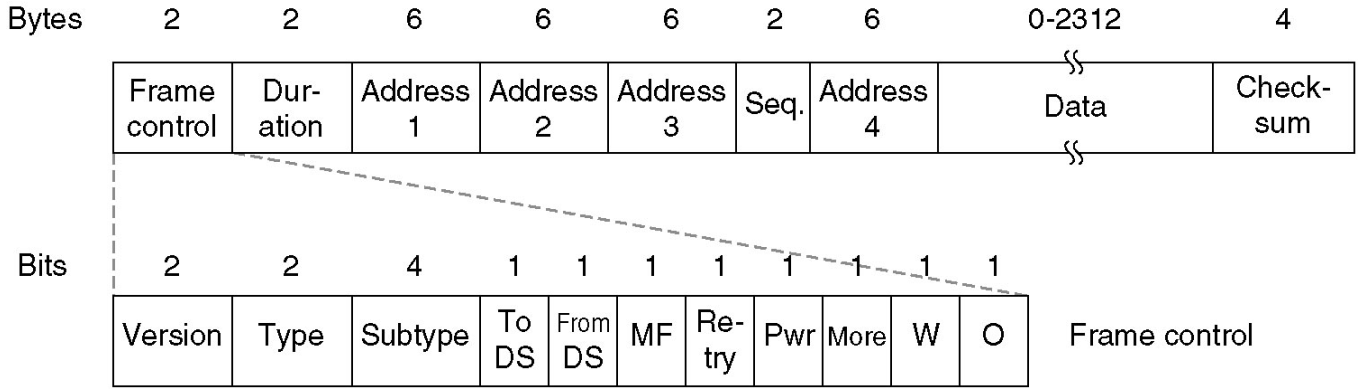
NAV: Network Allocation Vector

IEEE 802.11: FRAME STRUCTURE



- Type: data, control, management
- Subtype: RTS, CTS, etc.
- To DS/From DS: to or from intercell distribution system (e.g. Ethernet)
- MF: more fragment will follow
- Retry: this is a reTx.
- Pwr: used by BS to direct MS into sleep or out of sleep
- More: sender has more frames to the receiver
- W: body has been encrypted by WEP (Wired Equivalent Privacy)
- O: frames with this bit on must be processed in order.

IEEE 802.11: FRAME STRUCTURE



- Duration: how long this frame and its Ack will occupy the channel.
- 4 Addresses: 1 for source Host, 1 for destination Host
  - The other two are for intercell frames: source/destination BS
- Seq: sequence number of fragments (12-bit identify frame and 4-bit identify fragment)
- Data: payload, up to 2312 bytes
- Checksum
- Management and Control frames are shorter.

OUTLINE

- Functions of MAC Sublayer
- MAC Protocols
- Ethernet (IEEE 802.3)
- Data Link Layer Switching
- Wireless LAN
- **Broadband Wireless: WiMax**
- Bluetooth

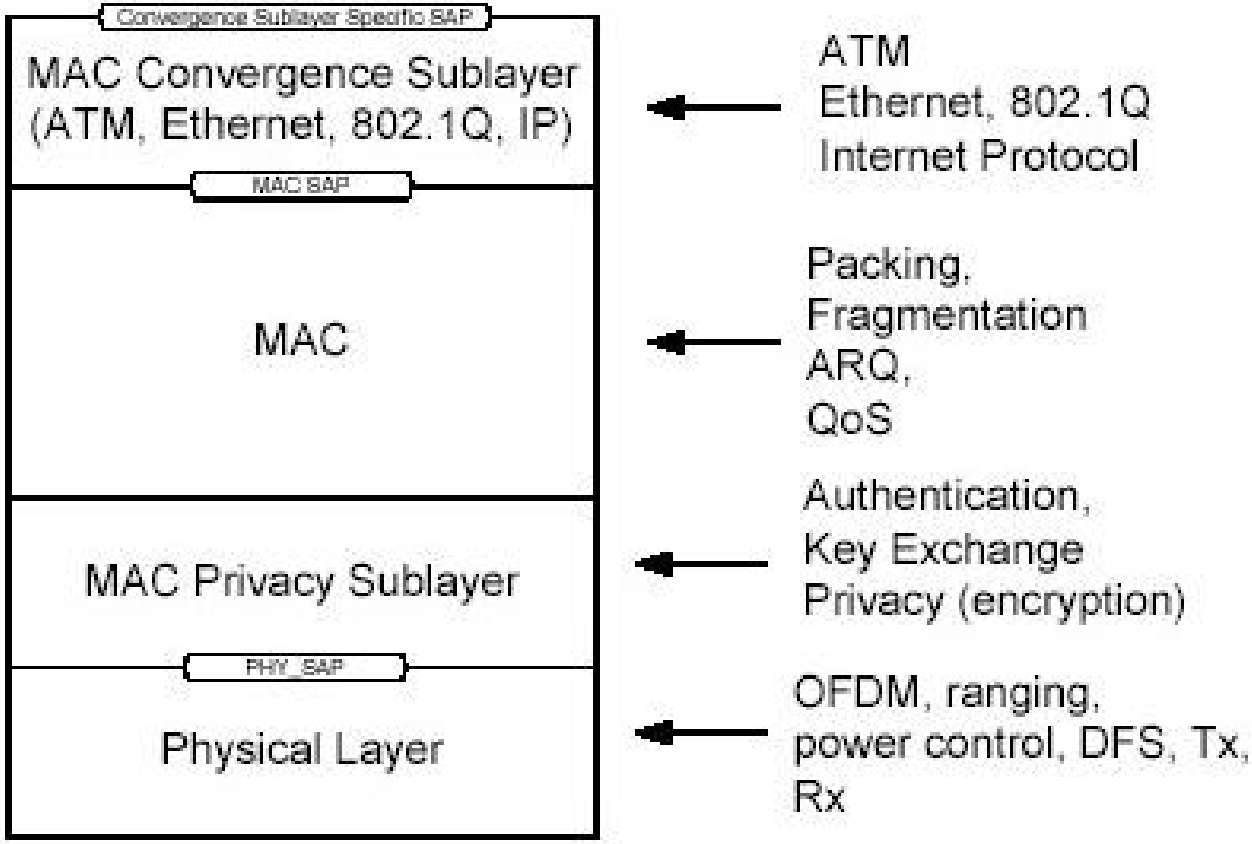
WIMAX

- **What is WiMAX?**
  - Worldwide Interoperability for Microwave Access.
  - A standard based technology that provides broadband wireless connections over long distances ➔ Wireless MAN
- **Standard History**
  - IEEE 802.16 (Dec. 2001) (no WiMax yet)
    - Line of sight (LOS) only
  - IEEE 802.16a (Jan. 2003)
    - Non-line of sight, point-to-multi-point applications.
    - Focused on out-of-door fixed services.
  - IEEE 802.16-2004 (Also called 802.16REVd) (3rd Quarter of 2004)
    - Enhancement of 802.16a, provide service to indoor users
  - IEEE 802.16e (Dec. 2005)
    - Support mobile users

WIMAX: STANDARDS COMPARISON

	802.16	802.16a/ 802.16REVd	802.16e
Spectrum	10-66GHz	<11GHz	<6GHz
LOS?	Yes	No	No
Data rate	32-134Mbps	Up to 75Mbps	Up to 15Mbps
Bandwidth	20, 25 and 28 MHz	1.25MHz ~ 20MHz	1.25MHz ~ 20MHz
Modulation	QPSK, 16QAM, 64QAM	OFDMA QPSK, 16QAM, 64QAM	Scalable OFDMA
Mobility	Fixed	Fixed	Mobile
Cell Radius	1 ~ 3 miles	3 to 5 miles Could be 30 miles	1 ~ 3 miles

WIMAX: PROTOCOL STACK



WIMAX: PHYSICAL LAYER

- **Two duplex mode**
  - TDD: Time division duplex
  - FDD: frequency division duplex
  - (Different from TDMA and FDMA!)
- **Support three kinds of physical layer technology**
  - Single carrier (one channel)
  - OFDM with 256 carriers (IEEE 802.16-2004)
    - 55 (28 low and 27 high) subcarriers are used for guard band
    - 8 subcarriers are used for pilot signal (for channel estimation)
    - 192 subcarriers are for users (192 simultaneous data streams!)
  - OFDM with 2048 carriers

WIMAX: MAC SUBLAYER

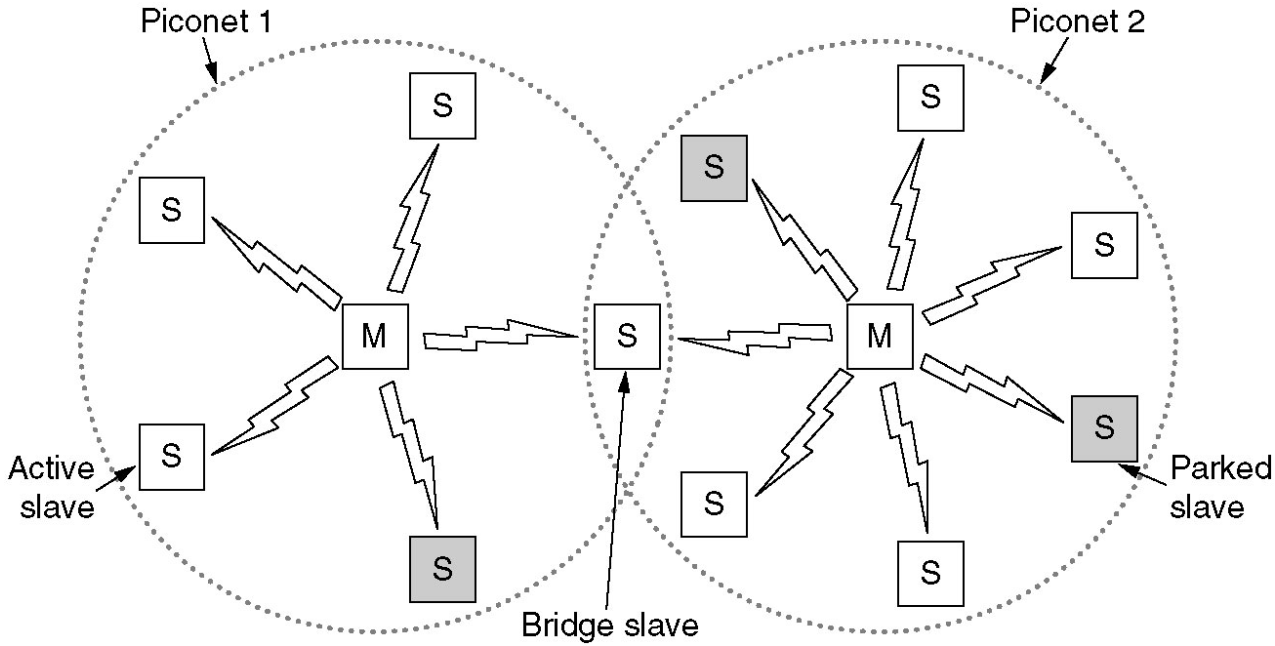
- **MAC**
  - Times are divided into slots and frames
  - Variable frame length
  - Scheduler at BS to coordinate the allocation of the channel
  - ARQ (Automatic Repeat Request) is used.
- **Services**
  - WiMax MAC provides connection oriented services to upper layer.
  - Four kinds of services
    - Constant bit rate
    - Real time variable bit rate
    - Non-real-time variable bit rate
    - Best effort
  - QoS can be maintained by the MAC sublayer
- **MAC Privacy Sublayer**
  - Authentication, encryption
- **MAC Convergence Sublayer**
  - Enable service can be offered to various upper layer protocols
    - E.g. IPv4, IPv6, ATM, etc.

OUTLINE

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- **Bluetooth**

BLUETOOTH

- **Bluetooth**
  - A wireless standard for interconnecting computing and communication devices using short-range, low power, inexpensive wireless radios.
  - Has been standardized as IEEE 802.15 (PAN).
- **Architecture: piconet, master, slave, bridge slave**



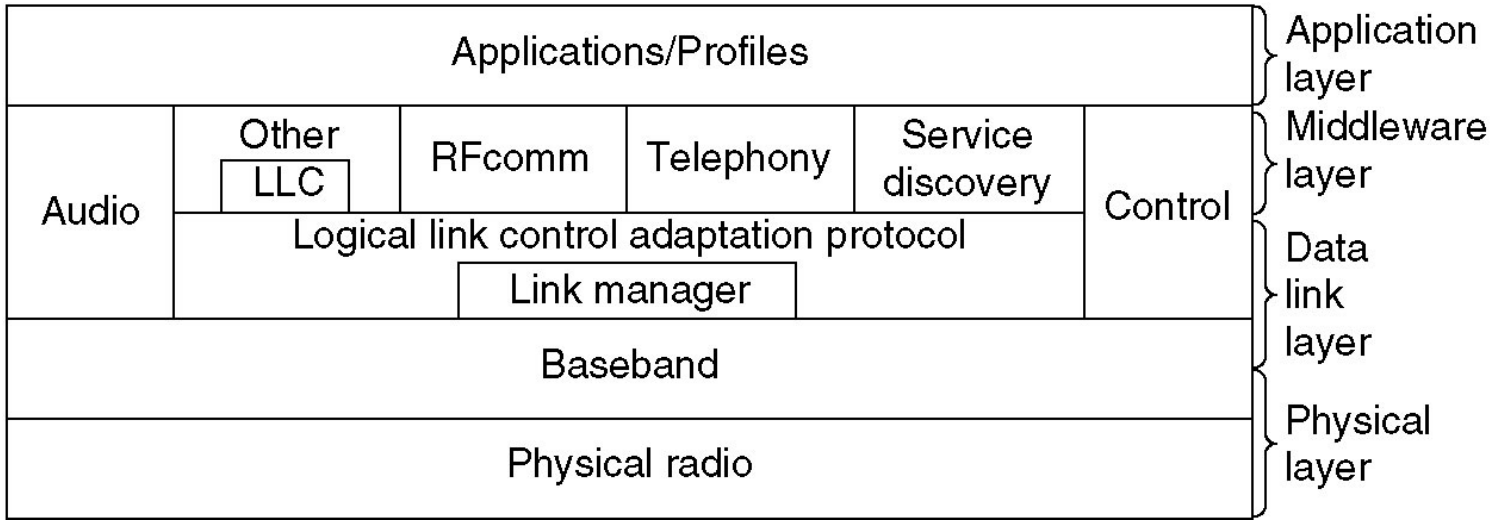
BLUETOOTH: APPLICATIONS

- **Applications (Bluetooth profiles)**

Name	Description
Generic access	Procedures for link management
Service discovery	Protocol for discovering offered services
Serial port	Replacement for a serial port cable
Generic object exchange	Defines client-server relationship for object movement
LAN access	Protocol between a mobile computer and a fixed LAN
Dial-up networking	Allows a notebook computer to call via a mobile phone
Fax	Allows a mobile fax machine to talk to a mobile phone
Cordless telephony	Connects a handset and its local base station
Intercom	Digital walkie-talkie
Headset	Intended for hands-free voice communication
Object push	Provides a way to exchange simple objects
File transfer	Provides a more general file transfer facility
Synchronization	Permits a PDA to synchronize with another computer

BLOOTH: PROTOCOL STACK

- **Protocol stack**

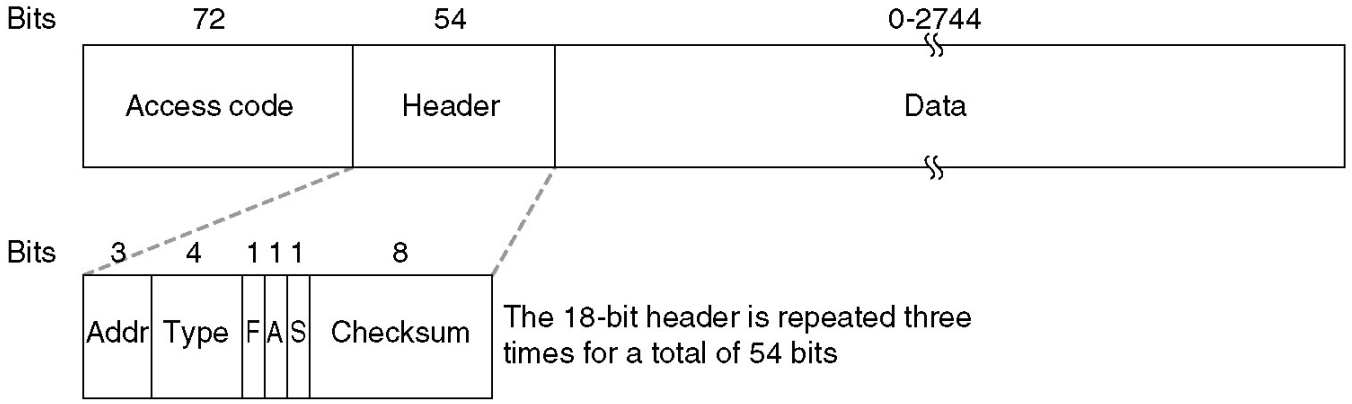


  - Physical radio:
    - FHSS; operation range: 10 meters; 2.4GHz ISM band.
    - Ultra Wide band (UWB) is under development
  - Baseband layer: analogous to MAC, also includes some elements of PHY (Physical) layer
  - Link manager: establishment of logical channels
  - RFcomm: emulate serial port on PC.

BLUETOOTH: BASEBAND LAYER

- **Functions of Baseband Layer**
  - Framing
  - 625 micro-second time slot.
    - Even slots: master; odd slots: slave ➔ TDMA with TDD
  - Two kinds of logic channels (links)
    - ACL (Asynchronous Connection-Less)
      - Packet-switched, best effort service
    - SCO (Synchronous Connection Oriented)
      - Real time data (e.g. telephony)
      - Fixed slots are reserved for such kind of link.

BLUETOOTH: FRAME STRUCTURE



- Access code: identifies the master
- Addr: identifies one of the 8 active slaves.
- Type: ACL, SCO, etc.
- F: flow bit. Slave tells master that its buffer is full ➔ flow control.
- A: acknowledgement bit. Piggyback Ack.
- S: sequence number. Window size?
- The 18-bit header is repeated 3 times to form the 54-bit header ➔ repetition code with error correction!

SUMMARY

- **MAC**
  - Determine how to allocate a single broadcast channel among multiple users.
- **MAC Protocols**
  - Contention based: ALOHA, CSMA, CSMA/CD
  - Non-contention based: Bitmap, Binary countdown
  - Limited contention: Adpative Tree Walk
  - WDMA, Wireless LAN
- **Ethernet (IEEE 802.3)**
  - Manchester Encoding
  - network size, data rate, and minimum frame size
  - Fast Ethernet, Gigabit Ethernet
- **Data Link Layer Switching**
  - Repeater, hub, switch, bridge, router, gateway
  - Destination table with backward learning
  - Spanning tree, VLAN
- **Wireless LAN**
  - MACA (RTS, CTS)
  - Physical Layer, MAC layer
- **Broadband Wireless: WiMax**
  - Standard history, physical layer, MAC layer
- **Bluetooth**
  - Architecture, frame structure